STAT 301 Worksheet 8 Name CHARLES SANDAHL

1. Find the covariance matrix for

$$B = \begin{matrix} 3 & 4 \\ 0 & 2 \end{matrix}$$

$$A = x - \frac{\mathbb{1} \cdot x}{n}$$

$$\begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix} - \frac{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}}{2} = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 1.5 & 3 \\ 1.5 & 3 \end{bmatrix} = \begin{bmatrix} 1.5 & 1 \\ -1.5 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1.5 & -1.5 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1.5 & 1 \\ -1.5 & -1 \end{bmatrix} = \begin{matrix} 1.5 \cdot 1.5 + -1.5 \cdot -1.5 & 1.5 \cdot 1 + -1.5 \cdot -1 \\ 1 \cdot 1.5 + -1 \cdot -1.5 & 1 \cdot 1 + -1 \cdot -1 \end{matrix}$$

$$\begin{bmatrix} 4.5 & 3 \\ 3 & 2 \end{bmatrix}$$

2.Find the derivative of the following functions

a) $g(x) = 3x^2 + 5x + 1$

$g'(x) = 3 \cdot 2x^{2-1} + 5 \cdot 1x^{1-1} + 0$

$6x + 5$

b) $f(x) = \frac{e^x}{5}$

$f'(x) = \frac{e^x}{5}$

c) $h(x) = e^x(x^2 + 1)$

$h'(x) = e^{x'} \cdot (x^2 + 1) + e^x(x^2 + 1)'$

$e^x(2 \cdot 1x^{2-1} + 0)$

$e^x(2x)$

$e^x x^2 + e^x + e^x 2x$

$6x^2 + 6x - 36$

$2 \cdot 3x^{3-1} + 3 \cdot 2x^{2-1} - 36 \cdot 1x^{1-1}$

3. If $f(x) = 2x^3 + 3x^2 - 36x$,

a) find the intervals where $f(x)$ is increasing and the intervals where $f(x)$ is decreasing

$6(x^2 + x - 6)$

$6(x+3)(x-2)$

$x = -3 \quad x = 2$

$(-\infty, -3)$ = positive

$(-3, 2)$ = negative

$(2, \infty)$ = positive

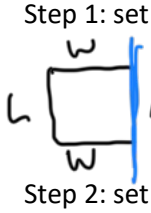
b) find the local minimum and maximum points of $f(x)$

local min = 2

local max = -3

4. A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

Step 1: set up the optimization function



$A = LW$

Step 2: set up the constrain and rewrite one variable using the other

$2400 = 2w\,L \qquad L = 2400 - 2w$

Step 3: substituting the optimization function and find the maximum area.

$A = (2400 - 2w)(w)$

$2400w - 2w^2$

$2400 - 4w = 0$

$600 = w$

$L = 2400 - 2(600)$

$L = 2400 - 1200$

$1200 = L$